Chapter 2
Evaluating Predictive Accuracy of Prognostic Model for Censored Time-to-Event Data Analysis in Clinical Trials

Kassu Mehari Beyene and Ding-Geng Chen

2.1 Introduction

Predictive models are collections of statistical techniques that aimed at determining the mathematical relationship between an outcome variable and a number of predictor variables with the objective of predicting future outcome. Regression analysis, decision trees, survival analysis, and other statistical and machine learning techniques are all types of predictive models. Predictive models are increasingly being used by organizations across various industries to make prediction in order to improve decision-making. In medicine, for example, predictive models can be used to predict the probability of either having (diagnostic model) or developing a particular disease or outcome (prognostic model). For physician, this is crucial because it helps to make evidence-based clinical decisions, optimize resource allocation, and deliver personalized care to patients. Prediction models are typically developed using regression techniques, such as logistic regression for diagnostic predictions and Cox proportional hazards regression for prognostic predictions. These regression models yield equations that allow the estimation of an individual's risk based on their values for one or more predictors, for example, their age, biomarkers, or genetic information. Several well-known examples of risk prediction models are commonly used in clinical practice. To predict the 10-year risk of

K. M. Beyene (✉)
College of Health Solution, Arizona State University, Phoenix, AZ, USA

Barrow Neurological Institute, Phoenix, AZ, USA
e-mail: kassu.beyene@barrowneuro.org

D.-G. Chen
College of Health Solution, Arizona State University, Phoenix, AZ, USA

Department of Statistics, University of Pretoria, Pretoria, South Africa
e-mail: Ding-Geng.Chen@asu.edu

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developing cardiovascular disease, for instance, many risk predictions tools based on survival models were developed and made publicly available during the last three decades; see, for example, (Damen et al., 2016). The well-known Framingham risk score combines risk factors such as age, blood pressure, smoking status, diabetes, high-density lipoproteins, and total cholesterol levels in order to calculate an individual's risk estimate.

It is important to note that the accuracy of predict model plays an important role in the quality of the decisions one make based on its risk estimates. Hence, it is extremely important to evaluate the accuracy of predictive models before using them for clinical decision-making. Calibration and discrimination of predictive models are two important components of evaluation of predictive performance. Calibration refers to how well the predicted probabilities from the model agree with observed outcome, while discrimination refers to how well the model discriminates between patients with the outcome of interest and those without it. Calibration curves, which are also called reliability diagrams (Wilks, 1990), are commonly used to measure model calibration. Discrimination is often assessed through metrics such as sensitivity, specificity, the Receiver Operating Characteristic (ROC) curve or the area under the ROC curve (AUC). In this chapter, we focus on the receiver operating characteristic curves and their associated summary measures.

2.2 Overview of ROC Curve and Its Associated Summary Measures

ROC curves have a long history in the field of radar technology and signal detection theory. Its origins can be traced back to World War II, when engineers and statisticians introduced to assess the radar receiver operators ability to accurately identify signal and noise in radar signals (Green et al., 1966; Egan, 1975). The ROC curve found applications in medical diagnostics in the 1960s, especially for evaluating diagnostic tests (Lusted, 1960). The concept gained further prominence in the 1970s and 1980s with its adoption in the field of machine learning, psychology, and radiology (Swets, 1973; Swets & Pickett, 1982; Metz, 1989; Spackman, 1989). The use of ROC curves in determining model accuracy is prevalent today in almost every field.

The ROC curve was originally developed to evaluate how well a test that produces a continuous score predicts a binary outcome (e.g., presence or absence of the event of interest). In addition to assessing the predictive performance of a model, the ROC curve is also used to select an optimal cutoff value for decision-making. The ROC curve is a graphical tool that plots the True Positive Rate (sensitivity) against the False Positive Rate (1-specificity) across different discrimination thresholds. Sensitivity measures the proportion of true positives correctly identified, while specificity measures the accuracy of correctly identifying true negatives. Given a continuous score M and binary outcome D with 1 indicating presence and 0 absence of an event, the true positive rate (TPR) and false positive

rate (FPR) can be written as

$$\mathrm{TPR}(m) = P(M > m|D = 1), \tag{2.1}$$
$$\mathrm{FPR}(m) = P(M > m|D = 0),$$

where m is a thresholds value. The ROC curve, which is defined as a plot of TPR versus FPR for all possible classification threshold values, in terms of function, can be written as

$$\mathrm{ROC}(u) = \mathrm{TPR}\{\mathrm{FPR}^{-1}(u)\}, \tag{2.2}$$

where $u \in [0, 1]$ and $\mathrm{FPR}^{-1}(u) = m$ is a threshold value such that $u = \mathrm{FPR}(m)$. The ROC curve is a standard technique for evaluating the predictive accuracy of a model over a range of trade-offs between TPR and FPR. In addition to assessing the predictive performance of a model, the ROC curve is used to select an optimal cutoff value for decision-making. Ideally, the ROC curve of a perfect model would be the one that rises steeply toward the upper-left corner (i.e., FPR = 0, TPR = 1), while an uninformative marker has an ROC curve that is the diagonal line (i.e., TPR = FPR for all threshold values).

Although the ROC curve provides most of the information about the accuracy of a marker, producing quantitative summary measures of the ROC curve is also useful because it provides valuable insights. Summarizing the ROC curve with a single numeric value has some important uses, such as providing a concise way of communicating the diagnostic performance, facilitating comparisons, and serving as the basis of inferential statistics for comparing ROC curves (Pepe, 2003). The two most commonly used ROC curve summary measures by far are the area under the ROC curve and the Youden index.

The AUC is the most widely used quantitative summary measure of the overall marker ability to classify individuals with and without the event of interest, and it can be written as

$$\mathrm{AUC} = \int_0^1 \mathrm{ROC}(u)du.$$

AUC values range from 0.5 to 1 and marker with higher AUC value would be preferable. For a marker with perfect classification, the AUC will be 1. A value of 0.5 for AUC indicates that the ROC curve will fall on the diagonal line and hence suggests that the diagnostic marker's discriminatory ability is not better than random guessing. Due to its meaningful interpretation, AUC is more interesting than other summary measures. The AUC can be interpreted as the probability that a randomly selected subject with the event has larger marker value than a randomly chosen subject without the event Hanley and McNeil (1982). The other interpretation is the average value of sensitivity (specificity) for all possible values of specificity(sensitivity) (Pepe, 2003; Zhou et al., 2011).

The Youden index (Youden, 1950) is another summary measure of the ROC curve for the accuracy of a diagnostic test. It can be defined as the maximum difference between the TPR and FPR. The cutoff value that corresponds to this maximum difference is known as optimal cutoff value. In clinical decision-making, the optimal cutoff value is useful to classify subjects according to the marker value (less than cutoff = healthy, greater than cutoff = diseased). The Youden index (J) can be written as

$$\mathrm{J} = \max_{m}\{\mathrm{TPR}(m) - \mathrm{FPR}(m)\}.$$

More about this and other optimal cutoff value selection methods is available in the literature; see, for example, Sima and Gönen (2013), Liu and Jin (2015), and Rota et al. (2015).

2.3 Definitions of Time-Dependent Predictive Accuracy Measures

Time-to-event survival data refer to a class of observational data commonly encountered in medical and epidemiological research, where the primary outcome of interest is the time until a specific event occurs. This event could be anything from the onset of a disease, disease recurrence, or death due to certain disease. The analysis of time-to-event data is a fundamental component of survival analysis, focusing on the duration until an event of interest takes place. Several statistical approaches including the widely used Cox proportional hazards model have been used for analyzing time-to-event data, providing insights into the probability of developing the event over time and identifying factors that may influence the time to the event. The time dependency of the event status of subjects, just like modeling, also causes complication to compute the ROC curve and its associated summary measures using simple proportions. Therefore, in literature, various extensions of ROC curve definitions for the case of time-dependent event have been proposed. These definitions are presented in upcoming subsections.

2.3.1 Incident/Dynamic

The incident true positive rate (TPR) and dynamic false positive rate (FPR) are determined by categorizing the risk set at time t into two groups: those who experience the event (cases) and those who have not experienced the event yet (controls). In this definition, incidence cases are defined as individuals who have experienced the event at a particular time point (i.e., $T = t$), and the dynamic controls are individuals who will experience an event after the time point (i.e., $T > t$) (Heagerty & Zheng, 2005). In this context, some individuals do not fall

into either the control or case group, specifically when their event time (T) is less than the target time (t), i.e., $T < t$. Using these definitions of event status, the time-dependent definition for the incident TPR (TPR_t^I) and dynamic FPR (TPR_t^D) can be expressed as follows:

$$\begin{aligned} \mathrm{TPR}_t^I(m) &= P(M > m | T = t), \\ \mathrm{FPR}_t^D(m) &= P(M > m | T > t), \end{aligned} \tag{2.3}$$

where $m \in \mathbb{R}$ is a fixed cutoff value.

The time-dependent ROC curve corresponding to this definition ($\mathrm{ROC}_t^{I/D}$) can defined as the plot of TPR_t^D versus FPR_t^I for all possible cutoff values m. Alternatively, it can be defined as

$$\mathrm{ROC}_t^{I/D}(u) = \mathrm{TPR}_t^I \left\{ \left[\mathrm{FPR}_t^D \right]^{-1} (u) \right\}, \tag{2.4}$$

where $u \in [0, 1]$ and $m = \mathrm{FPR}_t^{-1}(u)$ is the threshold value such that $u = \mathrm{FPR}_t(m)$.

2.3.2 Incident/Static

Etzioni et al. (1999) and Slate and Turnbull (2000) introduced an alternative time-dependent TPR and FPR definition known as incident TPR and static FPR. In this context, incident cases are identified as individuals experienced an event of interest at time t, whereas static controls are defined as subjects who remain event-free during a predetermined follow-up period, i.e., $T > t*$. Based on this, incidence TPR (FPR_t^I) and static FPR (FPR_{t*}^S) are defined as

$$\begin{aligned} \mathrm{TPR}_t^I(m) &= P(M > m | T = t), \\ \mathrm{FPR}_{t*}^S(m) &= P(M > m | T > t^*). \end{aligned} \tag{2.5}$$

As per this definition, the time-dependent ROC curve and the AUC can be defined accordingly.

2.3.3 Cumulative/Dynamic

The widely used extension of the definitions of the TPR and the FPR in the context of time-to-event data is the one introduced by Heagerty et al. (2000). The essence of their approach involves classifying the outcome at a specific prediction time of interest, denoted as t. Heagerty et al. (2000) defined "cumulative cases" (C) as individuals who experience the event of interest before time t and "dynamic

controls" (D) as those who remain event-free until time t. Using this classifications, the cumulative True Positive Rate (TPR_t^C) and dynamic False Positive Rate (FPR_t^D) at the fixed time t, respectively, can be written as follows:

$$\begin{aligned} \text{TPR}_t^C(m) &= P(M > m | T \leq t), \\ \text{FPR}_t^D(m) &= P(M > m | T > t), \end{aligned} \tag{2.6}$$

where $m \in \mathbb{R}$ is a fixed cutoff value.

These definitions are particularly suitable when the primary scientific interest involves classifying between individuals who experience the event before a specific time t and those who remain event-free beyond that time. This particular classification has greater clinical importance compared to the other definitions (Blanche et al., 2013); hence, in this chapter we focus on the cumulative/dynamic definition.

The time-dependent AUC corresponding to the time-dependent ROC curve can be written as

$$\text{AUC}_t = \int_0^1 \text{ROC}_t(u)du,$$

where $\text{ROC}_t(.)$ is any of the previously defined time-dependent ROC functions.

The time-dependent Youden index, which corresponds to the time-dependent TPR and FPR defined earlier, can be written as

$$\text{J}_t = \max_m \{\text{TPR}_t(m) - \text{FPR}_t(m)\},$$

where $\text{TPR}_t(.)$ and $\text{FPR}_t(.)$ are the time-dependent TPR and FPR, respectively.

2.4 Cumulative/Dynamic ROC Estimator for Right-Censored Data

In the context of time-to-event data with right censoring, various methods have been introduced to estimate the time-dependent ROC curve and its associated AUC, considering the specific challenges posed by right censoring; see, for example, Heagerty et al. (2000), Martínez-Camblor et al. (2016), Heagerty and Zheng (2005), Martínez-Camblor and Pardo-Fernández (2018), Li et al. (2018), Chambless and Diao (2006), Beyene et al. (2019), and Beyene and Chen (2024) and the references given therein. In this section, we review the recently introduced time-dependent ROC curve and the associated AUC estimators of Beyene and El Ghouch (2020).

In their work, Beyene and El Ghouch (2020) presented a novel non-parametric smoothed estimator for time-dependent ROC curves and the associated AUC. This method involved defining the ROC curve as a weighted distribution function. In their approach, the authors employed an imputation technique to estimate the event

status for censored subjects with unknown outcomes. This imputation concept was initially introduced in the time-dependent ROC curve estimation methods proposed by Martínez-Camblor et al. (2016) and Li et al. (2018).

Let us assume that the observed dataset consists of $\{(Y_i, \Delta_i, M_i), i = 1, 2 \ldots, n\}$ where $Y_i = \min(T_i, C_i)$ is the observed survival time, T_i is the time to the event of interest, C_i is the censoring time, $\Delta_i = I(T_i \leq C_i)$ is the censoring indicator and M_i is a continuous baseline marker of the ith subject. Moreover, let us assume that T and C are conditionally independent given the marker M. With some algebra, Beyene and El Ghouch (2020) derived an empirical time-dependent ROC curve estimator as

$$\widehat{\mathrm{ROC}}_{emp}(u) = \frac{\sum_{i=1}^{n} \hat{W}_{ti} I(\hat{Z}_i \leq u)}{\sum_{i=1}^{n} \hat{W}_{ti}}, \tag{2.7}$$

where $\hat{W}_t = \hat{P}(T \leq t)|Y, \Delta, M) = \left[1-(1-\Delta)\frac{\hat{S}(t|M)}{\hat{S}(Y|M)}\right] I(Y \leq t)$ is the event status of the subjects that have the value 1(0) for case(control) and $\frac{\hat{S}(t|M)}{\hat{S}(Y|M)}$ for censored subject with unknown event status, and $\hat{Z}_i = \widehat{FPR}_t(M_i)$ is the time-dependent FPR estimator that can be given by

$$\widehat{FPR}_t(m) = \frac{\sum_{i=1}^{n} (1 - \hat{W}_{ti}) I(M_i \geq m)}{\sum_{i=1}^{n} (1 - \hat{W}_{ti})} \tag{2.8}$$

Regarding the estimation of the required conditional survival distribution function $S(.|M)$ for the calculation of weight W_t, the authors proposed to use the Beran estimator, which can be expressed as:

$$\hat{S}(t|M = m) = \prod_{i=1}^{n} \left[1 - \frac{I(Y_i < t)\Delta_i \nu_i(m)}{\sum_{j=1}^{n} I(Y_j \geq Y_i)\nu_j(m)} \right] \tag{2.9}$$

where $\nu_i(m)$ is the Nadaraya–Watson weight, which can be written as

$$\nu_i(m) = \frac{K_b(m - M_i)}{\sum_{j=1}^{n} K_b(m - M_j)}$$

where $K_b(.) = K(./b)$ with K is a kernel smoothing function and $b \equiv b_n$ is the bandwidth.

The empirical time-dependent ROC curve estimator given in eq. (2.7) is smoothed using kernel smoothing technique. The smoothed time-dependent ROC

curve estimator is written as

$$\widehat{\mathrm{ROC}}_{smo}(u) = n^{-1} \sum_{i=1}^{n} \hat{\mathcal{W}}_i K \left(\frac{Q(u) - Q(\hat{Z}_i)}{h} \right). \tag{2.10}$$

where $\hat{\mathcal{W}}_i = \hat{W}_i / n^{-1} \sum_{i=1}^{n} \hat{W}_i$, h is the smoothing parameter and $Q(.)$ is a quantile transformation function that used to handle a boundary problem, which was caused due to the fact that the random variable $\hat{Z}$ has bounded range (i.e., $\hat{Z} \epsilon [0, 1]$). To select the optimal value of the smoothing parameter h, Beyene and El Ghouch (2020) introduced three data-driven approaches (i.e., the normal reference (NR), cross-validation (CV), and plug-in (PI)). Details about these approaches can obtained in their published work, see Beyene and El Ghouch (2020).

To estimate the time-dependent AUC, Beyene and El Ghouch (2020) introduced a simple empirical and a numerical integration approach. The simple estimator is given as

$$\widehat{AUC}_t = 1 - n^{-1} \sum_{i=1}^{n} \hat{\mathcal{W}}_i \hat{Z}_i. \tag{2.11}$$

To estimate standard error and make inference on the time-dependent AUC, a nonparametric bootstrap approach was proposed. Nonparametric bootstrap sampling is a resampling technique widely used in statistics to assess the sampling variability of statistical estimators without making assumptions about the underlying data distribution. This method involves creating numerous simulated datasets by randomly sampling observations with replacement from the original dataset. By performing calculations on each of these bootstrap samples, the variability and uncertainty in the estimator can be quantified. One common application of bootstrapping is the construction of percentile bootstrap confidence intervals, where the desired quantiles of the distribution of the estimator are extracted from the bootstrap sample distribution. To perform bootstrap sampling, follow these steps: (1) Draw a random sample with replacement from the original dataset to create a bootstrap sample. (2) Calculate the statistic of interest using the bootstrap sample. (3) Repeat steps 1 and 2 a large number of times (e.g., 1000) to create a distribution of the estimator. (4) Construct the confidence interval using the desired percentiles from the bootstrap distribution. Details about this procedure can obtained in Beyene and El Ghouch (2020).

2.4.1 Breast Cancer Clinical Data

These data are from a study conducted on a sequence of 295 breast cancer tumors in women that were extracted from the Netherlands Cancer Institute's fresh-frozen tissue bank. The criteria to include these patients were: primary invasive

breast carcinoma with diameter less than 5 cm; age at diagnosis less than 53 years; diagnosis year between 1984 and 1995; and no previous cancer except non-melanoma skin cancer. Treatment involved modified radical mastectomy or breast-conserving surgery, axillary lymph node dissection, and radiotherapy. Further details about the study can be found in Van De Vijver et al. (2002). This dataset is available in the R package `dynpred` as the `nki` data set (Putter, 2015). Among the 295 patients, 79 died by the end of the follow-up, and the remaining are censored subjects. All patients were assessed at least annually for a period of at least 5 years, and the median follow-up among these patients was 6.7 years. The main prognostic factors in breast cancer include age, tumor size, vascular invasion, histological grade, estrogen level, number of positive lymph nodes and hormone-receptor status. In this analysis, we assess the predictive accuracy of the quantitative prognostic factors age (M_1), tumor size (M_2), and the linear combination of age and tumor size ($M_3 = \hat{\beta}_1 * age + \hat{\beta}_2 * tumorsize$, where $\hat{\beta}_i, i = 1, 2$ are the regression estimates obtained form fitted Cox model). In order to assess how well these prognostic factors can predict outcomes, we calculate the time-dependent ROC curves and the corresponding AUC values.

In Table 2.1, the estimated time-dependent AUC curve for three markers ($M_1, M_2,$ and M_3) is displayed along with the associated bootstrap standard errors and 95% confidence intervals (CI) computed at prediction time $t = 5$ and $t = 10$. These values were obtained using the R package `cenROC`, and the reported standard errors and confidence intervals were generated from 1000 bootstrap samples. The prognostic factor age has smaller AUC value compared to both the tumor size and the linear combination of age and tumor size. Furthermore, unlike the other factors, the AUC value associated with age is not significantly different from 0.5. This is due to the fact that the confidence interval encompasses the null value ($AUC = 0.5$), implying a lack of sufficient evidence to conclude that age exhibits better discriminatory ability than random guessing. The predictive ability of the factors tumor size and the combination of age and tumor size is very similar. In addition, both of these factors exhibit statistically significant and good overall predictive ability. This is also evident in the estimated empirical and smoothed time-dependent ROC curves shown in Fig. 2.1.

Table 2.1 Estimated AUC values with 95% bootstrap confidence intervals for prognostic factors tumor size, age, and combined for $t = 5$, 10 years

Prognostic factor	Estimate	Standard error	95% bootstrap CI	
$t = 5$				
Tumor size	0.683	0.043	0.598	0.767
Age	0.574	0.053	0.477	0.685
Combined	0.677	0.044	0.584	0.759
$t = 10$				
Tumor size	0.637	0.041	0.561	0.723
Age	0.583	0.046	0.491	0.670
Combined	0.644	0.043	0.558	0.729

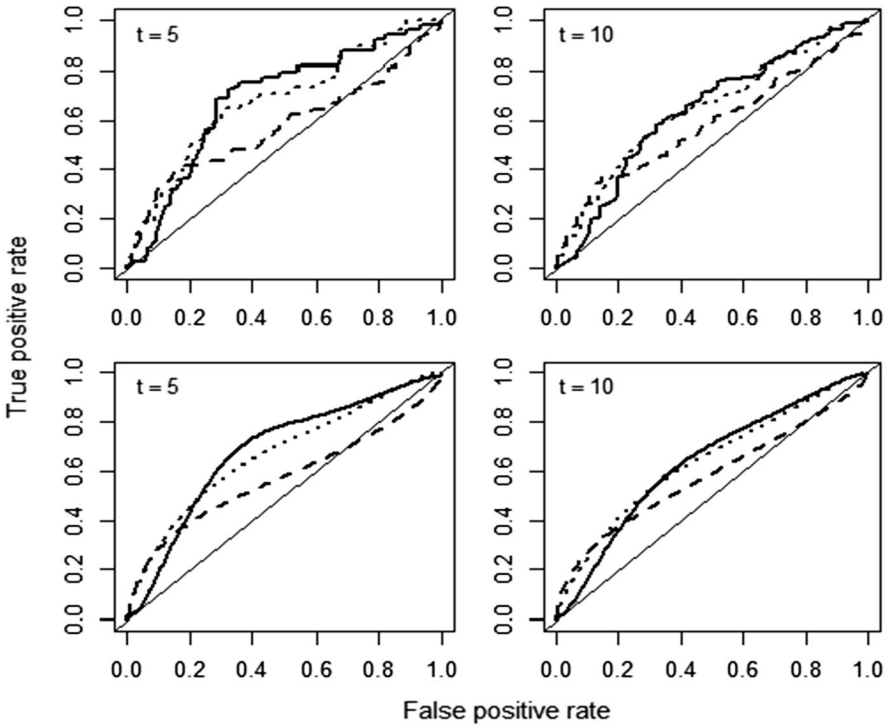

Fig. 2.1 Estimated time-dependent ROC curve from the empirical (top row), and smooth (bottom row) for prognostic factors tumor size (solid line), age (dashed line), and combined (dotted line). The prediction time considered is 5 and 10 years

2.5 Cumulative/Dynamic ROC Estimator for Interval-Censored Data

Interval-censored survival data refer to a specific type of time-to-event data in the field of survival analysis, where the exact event occurrence time is unknown but is known to have happened within lower and upper bounds. In contrast to right-censored data, where only a minimum of survival and censoring time is known, interval-censored data provide richer information about the time until an event of interest. This type of data commonly arises in follow-up clinical studies where follow-up intervals are defined, and events are only detected during these specific intervals. Analyzing interval-censored time-to-event data poses unique challenges in statistical modeling, requiring specialized techniques. This gives rise to difficulty to estimate the time-dependent ROC curve and its associated summary measures. To show why estimating time-dependent ROC curve of interval-censored data is difficult, we considered a hypothetical dataset with five subjects, see Fig. 2.2. This figure shows individuals' event status of the hypothetical dataset. From this figure,

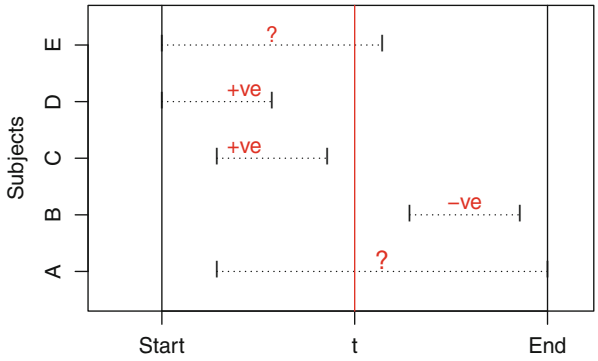

Fig. 2.2 Individuals event status for hypothetical dataset

individuals A and E have unkown event status because the considered prediction time t lies within the observed interval. Individuals C and D developed the event because the prediction time is larger than the upper bound of the two intervals. In contrast, individual B does not develop the event because the prediction time is smaller than the lower bound of the observed interval. Due to this, in literature, there is scarcity in methodologies for estimating the time-dependent ROC curve and its summary measures for interval-censored data, compared to the well-developed approaches for right-censored data. Some of the existing literature about time-dependent ROC curve and its associated summary measures for interval-censored data are Li and Ma (2011), Wu and Cook (2020), Díaz-Coto et al. (2020), and Beyene and El Ghouch (2022). In this section, we review and apply the methods introduced by Beyene and El Ghouch (2022). The time-dependent ROC curve and its associated summary measures estimation methods of interval-censored data proposed by Beyene and El Ghouch (2022) extend the approach they introduced for right-censored data which is given in Eqs. (2.7) and (2.10) for the empirical and smooth ROC curve, respectively, and Eq. (2.11) for the AUC. This extension modifies the weights of these equations to account for the interval-censored nature of the data. The weights in the time-dependent ROC curve and its associated summary measures estimators for interval-censored are given as

$$W_t = I(R \leq t) + \left\{ \frac{S(L|M) - S(t|M)}{S(L|M) - S(R|M)} \right\} I(L < t < R). \tag{2.12}$$

This weight formula exhibits a noteworthy property of reducing to established results under specific conditions. In the case of an exact observation, where the event time is precisely known (i.e., $L = T = R$), W_t simplifies to an indicator function (i.e., $I(T \leq t)$) and in the case of right-censored observation (i.e., $L < T < R = \infty$), W_t reduces to the weight given in Beyene and El Ghouch (2023), Li et al. (2018), and Martínez-Camblor et al. (2016). The weight function, W_t, given in Eq. (2.12), involves the conditional survival function, $S(.|M)$, which is unknown

and needs to be estimated from the observed data. Beyene and El Ghouch (2022) addressed this challenge by proposing three estimation methods: non-parametric (i.e., spline approach), semi-parametric (i.e., Cox proportional hazards model), and parametric survival models. In addition to the AUC, for interval-censored data, Beyene and El Ghouch (2022) also introduced the time-dependent Youden index estimator, which is given as

$$\hat{\mathtt{J}}_t(m) = \widehat{\mathtt{TPR}}_t(m) - \widehat{\mathtt{FPR}}_t(m). \tag{2.13}$$

The time-dependent ROC curve and its associated summary measures estimation methods introduced for the right- and interval-censored data presented in Sects. 2.4 and 2.5, respectively, are implemented in R package `cenROC` (Beyene and El Ghouch, 2023).

2.5.1 Age-Related Eye Disease Study (AREDS) Data

The Age-Related Eye Disease Study (AREDS) is a long-term multicenter, prospective study on persons aged 55–80 years. It is designed to assess the clinical course, prognosis, and risk factors of both AMD and cataract. In this study, each subject was followed every 6 months (in the first 6 years) or 1 year (after year 6) for about 12 years. To measure the disease progression, a severity score, scaled from 1 to 12 (with a more significant value indicating more severe AMD), was determined for each eye of each participant at every examination. The outcome of interest is the progression time-to-late-AMD of each eye, where time-to-late-AMD is defined as the time from the baseline visit to the first visit when the severity score reached 9 or above. Details about the study design can be obtained in Sun et al. (2019) and Sun and Ding (2021). In this example, we used data obtained from 629 Caucasian participants who had at least one eye in moderate AMD stage at baseline, defined by severity scores between 4 and 8. In addition to the outcome of interest, this dataset also contains two non-genetic prognostic risk factors, margin-specific baseline AMD severity score (SevScaleBL) and age at baseline (ENROLLAGE), and a genetic factor (rs2284665). This dataset is freely available in the R package `CopulaCenR` as the `AREDS` dataset, see Sun and Ding (2023) and Sun and Ding (2020). Measurements obtained from the two eyes of each individual are correlated, which violates the assumption of the method used, the analysis was done for each eye. This analysis is to assess the predictive accuracy of the composite prognostic risk score of each eye that defined as linear combination of risk factors: SevScaleBL, ENROLLAGE and rs2284665 (i.e., $M = \hat{\beta}_1 * SevScaleBL + \hat{\beta}_2 * ENROLLAGE + \hat{\beta}_3 * rs2284665$ where $\hat{\beta}_i, i = 1, 2, 3$ are the regression coefficient estimates obtained form fitted Cox model using the R package `icenReg`, see Anderson-Bergman (2017)). We also compare the predictive performance of the risk scores of the left and right eyes.

To compare the accuracy of the two risk scores defined earlier in predicting AMD disease progression of the two eyes, we estimated the time-dependent ROC

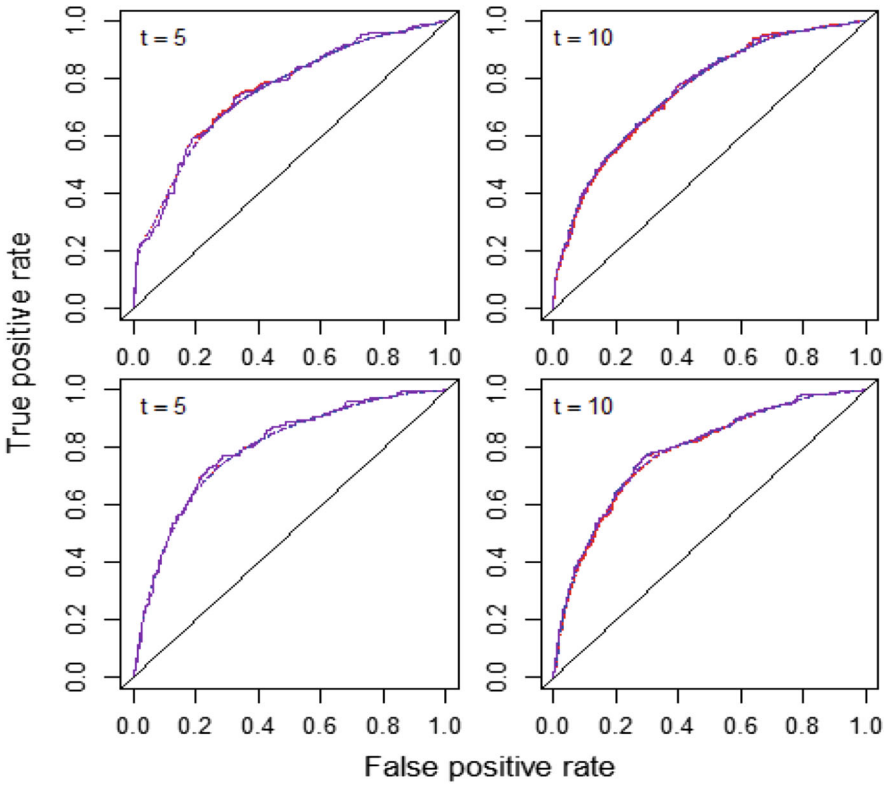

Fig. 2.3 Estimated time-dependent ROC curve for the left eye (top row) and right eye. The empirical estimates (dashed lines) using the non-parametric spline (red line), semi-parametric Cox (blue line), and parametric Weibull (purple line), and the smoothed estimates (solid lines) using the non-parametric spline (red line), semi-parametric Cox (blue line), and parametric Weibull (purple line). The prediction time considered is 5 and 10 years

curves, and the associated AUC and Youden index values at prediction time $t = 5$, and 10 years. The estimated time-dependent ROC curves (both smoothed and non-smoothed for each eye) using non-parametric (Spline), semi-parametric (Cox) and parametric (Weibull) approaches are presented in Fig. 2.3. This plot shows that the predictive ability of the each risk score is good, and their predictive performances are very similar. Furthermore, the three methods of estimating time-dependent ROC curve give very similar results. The results of time-dependent AUC estimation presented in Table 2.2 and the time-dependent Youden index in Table 2.3 also show similar performances.

Table 2.2 Estimated time-dependent AUC for each eye using the empirical non-parametric, semi-parametric, and parametric methods at $t = 5$ and 10 years

	Estimated time-dependent AUC		
t (in years)	Non-parametric	Semi-parametric	Parametric
Left eye			
5	0.7564	0.7557	0.7558
10	0.7599	0.7625	0.7617
Right eye			
5	0.7975	0.7967	0.7973
10	0.7846	0.7873	0.7883

Table 2.3 Estimated time-dependent Youden index for each eye using, both empirical and smoothed, non-parametric, semi-parametric, and parametric methods at $t = 5$ and 10 years

	Empirical time-dependent Youden index			Smoothed time-dependent Youden index		
t (in years)	Non-parametric	Semi-parametric	Parametric	Non-parametric	Semi-parametric	Parametric
Left eye						
5	0.4129	0.4106	0.4107	0.3906	0.3883	0.3887
10	0.3796	0.3824	0.3822	0.3622	0.3675	0.3663
Right eye						
5	0.4892	0.4877	0.4890	0.4609	0.4592	0.4606
10	0.4780	0.4794	0.4804	0.4490	0.4513	0.4516

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Dr. Kassu Mehari Beyene is currently a Clinical Biostatistician at Barrow Neurological Institute (BNI). Before joining BNI, he received his PhD in Biostatistics from Université catholique de Louvain at Louvain-la-Neuve and worked as a postdoctoral researcher at the College of Health Solutions, Arizona State University at Phoenix. After earning a PhD in Biostatistics from Catholic University of Louvain (Belgium). He also worked as an assistant professor in Biostatistics at the Department of Statistics, Wollo University, Ethiopia. His research interest includes the ROC curve and other related approaches for evaluating predictive accuracy of biomarkers, survival data analysis, infectious disease modeling, and longitudinal data analysis. He has published four papers, with another in press in ROC curve estimation for censored time-to-event data, as well as an R package for ROC curve estimation in the context of survival data. Recently, at the 79th Deming Conference held in Philadelphia from December 4 to 6, 2023, he presented a talk on ROC curves and related measures estimation for censored time-to-event data.

Dr. Ding-Geng Chen is a fellow of the American Statistical Association and is currently the executive director and professor in biostatistics at the College of Health Solutions, Arizona State University. He is also an extraordinary professor and the SARChI research chair in biostatistics at the University of Pretoria, an honorary professor at the University of KwaZulu-Natal, South Africa. Dr. Chen was a professor in Biostatistics at the University of North Carolina at Chapel Hill, a professor in biostatistics at the University of Rochester Medical School, and the Karl E. Peace Endowed Eminent Scholar Chair in Biostatistics at Georgia Southern University. He is a senior biostatistics consultant for biopharmaceuticals and government agencies with extensive expertise in biostatistics, clinical trials, and public health statistics and data sciences. Dr. Chen has more than 200 referred professional publications, coauthored/coedited 40 books on clinical trial methodology, meta-analysis, data science, causal inference, and public health research. Dr. Chen has been with the Deming Conference since 2006 as attendee first and as an invited speaker soon after that. He contributed significantly to the success of the Deming Conference as the Deming Publicity Chair, a tutorial moderator, and an invited speaker. As the most frequent speaker at the Deming Conference, Dr. Chen has been invited to 13 tutorials on different topics in biostatistics and was honored with the "Award of Recognition" from the Deming Conference Committee for highly successful advanced biostatistics workshop tutorials.